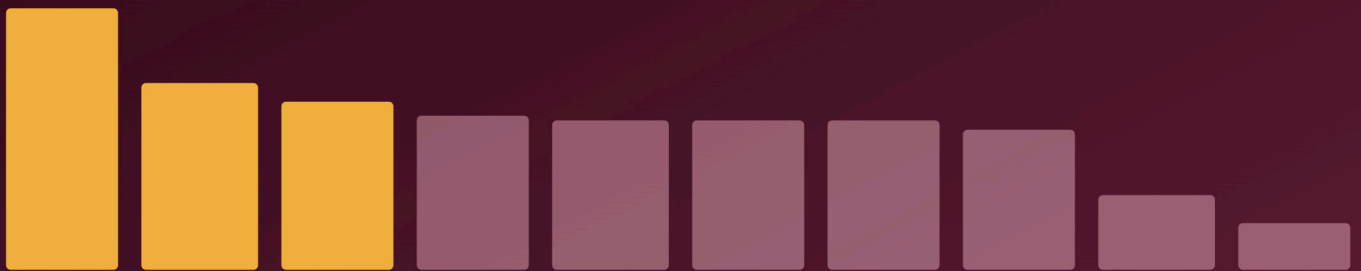


DEEP PYQ ANALYSIS

GATE EE 2027

High-Probability Question Bank
& 10 Full-Length Mock Tests



Marks per section · five counted papers

MD ANISH AHAMAD

GATE EE 2027 — Deep PYQ Analysis & High-Probability Question Bank

By MD ANISH AHAMAD · First Edition 2026

This preview contains 34 real, unedited pages from the 643-page book, exactly as they appear in the full PDF. Page numbers in the running heads and on the contents pages refer to the full book.

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Last-minute sheet (first pages)	pages 599–600	2
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The full book also has: all eleven subject chapters, 264 practice questions with worked solutions and four confidence figures each, all ten mock tests with keys and solutions, the hundred to work first, the what-not-to-skip chapter, the revision plan and the full last-minute sheet.

GATE EE 2027

Deep PYQ Analysis & High-Probability Question Bank

Five GATE Electrical Engineering papers, 2022 to 2026, counted question by question, a question bank in every subject chapter, and ten full-length mock tests built on what they show for the GATE Electrical Engineering paper

MD ANISH AHAMAD

First Edition · 2026

For the examination of February 2027

gateguide.co.in

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Preface

Every GATE book tells you what is in the syllabus. The syllabus is two pages long and the organisers publish it free. This book tries to answer the question the syllabus cannot: **of the eighty-five technical marks in the GATE Electrical Engineering paper, where do they actually land — and how much of that can anyone honestly claim to know?**

There is a reason that question is usually answered badly. The organisers have never published a marks breakdown, so every "subject-wise weightage" table you have seen is somebody's count of past papers, and usually it does not say where it put the questions that sit on a boundary between two sections. In a paper whose sections overlap as much as this one's do — a transfer function that belongs to Signals and Systems and to Control Systems, a diode rectifier that belongs to Analog Electronics and to Power Electronics, a synchronous machine that is a device in Electrical Machines and a source in Power Systems — that is not a small omission.

So this book does not use those tables. It counts the papers: the five most recent, all of them.

How this book was built

Five papers, counted question by question. Every GATE Electrical Engineering paper from 2022 to 2026 — one a year, 325 questions and 500 marks — was read one question at a time. Each question was assigned to a section of the 2027 syllabus and to a topic inside it, and its marks and question type were taken from that year's official answer key, never from the paper's layout. Each paper was made to close: 65 questions, 100 marks, 30 at one mark and 35 at two, General Aptitude at exactly 15. A count that does not close is a count with a mistake in it, and all five close.

Every citation names a year and a question number. Not once in this book will you find a claim like "this pattern keeps recurring". Electrical Engineering is set as one paper a year, so a citation such as "2025 (Q35)" names exactly one question, and every historical statement in this book can be checked against a paper you can download from an organising institute's own website.

Every boundary decision is written down. Forty-two of the 275 technical questions could honestly have been counted in a second section. Each records the section it was counted in, the section it could have gone to, and why, so that a reader who disagrees can move it and see exactly what changes. Thirty were not settled by any stated rule, and each of those is marked as counted at its lean rather than presented as certain. Section 4.7 lists them all.

The 2027 syllabus was compared with 2026, word by word, from the two official PDFs, by a program rather than by reading. The result is short: **no phrase added, no phrase removed, no phrase reworded.** The only change of wording is one corrected spelling — "Butterwoth" in 2026 is "Butterworth" in 2027. The 2024 and 2025 syllabi were compared with 2027 the same way and name the same phrases. Section 2 prints the comparison and the method. A change list that is wrong is worse than none, and a change list that says "nothing changed" is the one most likely to be doubted, so it is the one most worth generating from the source rather than asserting.

Every number in every question was computed, not remembered. Node and mesh equations were solved, phasors were added, transients and transfer functions were evaluated, per-unit and fault currents were worked through, converter averages and RMS values were integrated, and truth tables and counter sequences were enumerated. Every incorrect option is the output of a specific mistake — peak used for RMS, line voltage used for phase voltage, a sign slip in KVL — and the solution names that mistake, so that a candidate who chose it learns what they did rather than merely that they were wrong.

What this book refuses to do

It does not claim to know the paper. Four commitments follow:

- **Facts, analysis and predictions are labelled separately.** A statement about what happened in 2025 is marked FACT. A statement about what the data suggests is marked ANALYSIS. A statement about 2027 is marked PREDICTION. A statement that is the author's opinion is marked JUDGEMENT. They are never blended.
- **Confidence is reported in four dimensions, not one.** Being confident that an induction motor will be examined is not the same as being confident about a particular induction-motor question. Every question carries Topic, Concept, Question-pattern and Exact-question confidence, and the last is capped at 35 per cent because GATE repeats concepts, not wording.
- **The confidence figures are called estimates, because that is what they are.** They are not a fitted model and they have not been validated. Five papers could not validate them if anyone tried. Section 7 explains how they were arrived at and what they cannot support.
- **Absence of evidence is printed as absence of evidence.** Several topics named in the 2027 syllabus produced no question at all in five papers. Each is said so plainly, with no history invented to fill the gap.

Two things the counting found that should change how you study

Engineering Mathematics is the largest technical section in every one of the five papers. It carried 15, 13, 13, 15 and 18 marks — 74 of the 425 technical marks set, and more than any other section in each year. It also carries 10 of the 22 multiple-select questions the five papers set. A candidate who treats it as the section to fit in around the "real" electrical subjects is arranging the paper upside down.

Thirteen topics the syllabus names have never been the subject of a counted question. Among them are oscillators and feedback amplifiers, active filters, VCOs and timers, and Schmitt triggers with sample-and-hold circuits in Analog and Digital Electronics; oscilloscopes, digital voltmeters and phase, time and frequency measurement in Measurements; and series and shunt compensation and insulators in Power Systems. The 2024, 2025 and 2026 papers were set on syllabi that named every one of them, so for those three papers the zero cannot be explained away as "not on the syllabus then". That is not evidence that they will be skipped again — five papers are too few for that — but it is a fact that no previous-year practice can cover them, and this book writes questions on them on the authority of the syllabus alone and says so.

The two findings point the same way: the past papers are best used to learn how a topic is asked and roughly how much it carries, and the syllabus is the only guide to what else can be asked.

What is in it

The official 2027 syllabus with the 2026-to-2027 comparison generated from the source documents; a chapter on the organising institute and why none of the counted papers carries its name; the counted weightage of every section and, inside each section, of every topic, over all five papers; eleven subject chapters — one for each of the ten technical sections and one for General Aptitude — each with a question bank of newly written questions, every one with a worked solution and four confidence figures; a chapter explaining exactly what those figures mean and what they do not; a hundred-question priority list; a chapter on what not to skip; ten full-length mock tests in the 2027 pattern with keys and worked solutions, 650 questions more; a staged revision plan; a last-minute sheet; and a statement of sources and their limits.

Every question in the book is new. None is copied from a past paper, and the mock tests were written under a rule that forbids reusing any question from the subject chapters. What is borrowed from the past is the pattern, and the pattern is what repeats.

Section 1 — Executive Summary

If you read nothing else in this book, read this. Everything here is drawn from the five most recent official GATE Electrical Engineering papers — 2022 to 2026, one a year — counted question by question against their official answer keys, and from the official syllabus documents. Every figure can be checked; every claim names what it rests on.

Statements are labelled. **FACT** is something counted or quoted. **ANALYSIS** is an inference from it. **PREDICTION** is a statement about a paper that has not been set. They are never blended.

The eight things that matter most

1. The 2027 syllabus did not change a single examinable word

FACT. The 2026 and 2027 EE syllabus documents were compared word by word by a program. Result: **0 phrases added, 0 removed, 0 reworded**, across all ten sections and all 217 phrases. The only change of wording is one corrected spelling in Section 9: "Butterwoth" in 2026, "Butterworth" in 2027. Nine of the ten sections are identical word for word and punctuation mark for punctuation mark.

FACT. The 2024 and 2025 EE syllabi, compared with 2027 in the same way, also add, remove and reword no phrase. The 2025 text differs from 2027's only in that spelling; the 2024 text differs in that spelling and in fourteen changes of capitalisation or punctuation. The 2022 and 2023 syllabi could not be obtained from their organisers' sites, and this book says nothing about them.

ANALYSIS. This is the most useful fact in the book, and the least exciting. **At least the three most recent papers were set on the same named topics you will sit.** Nothing in that part of the record has to be discounted as deleted material, and no topic's absence from 2024, 2025 or 2026 can be excused as "not on the syllabus then".

What to do. Nothing you studied against the 2026 syllabus has been removed, and nothing new has been added that you must find time for. Verify the claim yourself: the two documents are two pages each.

2. Engineering Mathematics is the largest section in every paper

FACT. Engineering Mathematics carried **15, 13, 13, 15 and 18 marks** in 2022 to 2026: a mean of **14.8**, and **74 of the 425** technical marks set in five papers, 17.4 per cent. It was the largest technical section in all five papers. No other section reached 13 marks in any year.

FACT. Ten of the 22 multiple-select questions in the five papers were Engineering Mathematics questions.

ANALYSIS. This is the one ranking in the record that five papers have not disturbed. It is also a section whose methods — eigenvalues, Laplace-domain algebra, vector calculus, differential equations — decide questions in Signals and Systems, Control Systems and Electromagnetic Fields as well.

3. General Aptitude is 15 marks in every paper, and every question in it has been an MCQ

FACT. Ten questions — five at one mark, five at two — totalling exactly 15 marks, in all five papers. All fifty General Aptitude questions set in those papers were multiple-choice questions.

ANALYSIS. The only block of the paper with no variation at all. It is worth more than the five-paper mean of any technical section, Engineering Mathematics included.

What to do. Do not leave it to the last month. It is the most predictable 15 per cent of your score.

4. IIT Madras sets 2027, and none of the five counted papers is its paper

FACT. GATE 2027 is organised by IIT Madras. The five counted EE papers were organised by **IIT Kharagpur (2022), IIT Kanpur (2023), IISc Bengaluru (2024), IIT Roorkee (2025) and IIT Guwahati (2026).** IIT Madras last organised GATE in 2019, three years before the first paper this book counts.

ANALYSIS. There is no institute-matched evidence in this book's record. Any claim that "IIT Madras prefers" some kind of EE question is not an inference from anything this book counted. The one public decision IIT Madras has made about the EE paper is the syllabus — and it left every examinable word as it was.

5. Five papers show patterns, not trends

FACT. Summing the section-by-section differences in technical marks, consecutive papers differ by **20, 16, 16 and 24 marks** out of 85 (2022 to 2023, 2023 to 2024, 2024 to 2025, 2025 to 2026). Across all ten pairs of papers the difference runs from 12 marks (2022 against 2025) to 30 (2023 against 2026).

FACT. Not one of the ten technical sections rose in every step from 2022 to 2026, and not one fell in every step.

ANALYSIS. The paper's shape moves by 16 to 24 of its 85 technical marks from one year to the next, and there are only five years. Movements such as Signals and Systems falling from 9 marks in 2025 to 4 in 2026, or Power Electronics running 11, 6, 11, 10, 10, are single draws, not directions. **Nothing in this book is claimed as a trend.** Where a section moved sharply once, the book says so and says that five papers cannot tell that apart from chance.

What to do. Read the range beside every mean in this book. Your paper gets one value out of that range and you do not choose which.

6. Where the marks are

FACT. Marks by section in the five papers, counted against the 2027 syllabus, with the mean and range:

Section	2022	2023	2024	2025	2026	Mean	Range
1. Engineering Mathematics	15	13	13	15	18	14.8	13-18
General Aptitude	15	15	15	15	15	15.0	15-15
6. Power Systems	10	10	12	9	12	10.6	9-12
10. Power Electronics	11	6	11	10	10	9.6	6-11
4. Signals and Systems	9	11	11	9	4	8.8	4-11
2. Electric circuits	7	10	7	8	11	8.6	7-11
9. Analog and Digital Electronics	9	10	9	8	7	8.6	7-10
5. Electrical Machines	10	8	6	9	9	8.4	6-10
7. Control Systems	8	9	9	9	6	8.2	6-9
3. Electromagnetic Fields	5	4	4	3	6	4.4	3-6
8. Electrical and Electronic Measurements	1	2	3	5	2	2.6	1-5
Outside the 2027 syllabus	0	2	0	0	0	0.4	0-2

ANALYSIS. The top three technical sections — Engineering Mathematics, Power Systems and Power Electronics — carry 175 of the 425 technical marks, 35.0 a paper. Below them, six sections sit within a mark and a half of one another on the mean, from 9.6 down to 8.2, so their order is close to a tie. **Signals and Systems is the least predictable block in the paper:** it has run from 11 marks to 4. Power Systems, Control Systems, Analog and Digital Electronics and Electromagnetic Fields have the narrowest ranges, three marks each.

7. Most of the marks sit in topics examined four or five times — and thirteen named topics never

FACT. Grouping each section's syllabus into the topic groups Section 6 uses gives 84 groups. **10 appeared in all five papers, carrying 134 of the 425** technical marks; 12 appeared in four papers (99 marks); 20 in three (118 marks); 14 in two (47 marks); 15 in one (25 marks); and **13 in none**.

FACT. The thirteen are: inductance, magnetomotive force, reluctance and magnetic circuits (Section 3); the RMS and average value of a periodic waveform (Section 4); power generation concepts with AC and DC transmission concepts, series and shunt compensation, electric field distribution and insulators, and per-unit quantities (Section 6); digital voltmeters and multimeters, phase, time and frequency measurement, and oscilloscopes (Section 8); and oscillators and feedback amplifiers, active filters, VCOs and timers, and Schmitt triggers with sample-and-hold circuits (Section 9).

ANALYSIS. More than half the technical marks set so far, 233 of 425, sit in the 22 groups examined in four or five papers. That is the core to secure first. A zero means that no counted question was *about* the topic, as this book labelled them; it does not mean the idea never appeared inside another question — per-unit values and RMS values are used in several questions counted elsewhere. And a zero says nothing about 2027.

What to do. Study the four-and-five-paper topics to depth, and give the never-examined topics a short, honest reading: they are named, most are short, and no past paper can prepare you for them.

8. This is a paper of numerical answers, with a small and variable number of MSQs

FACT. MCQ, MSQ and NAT counts over the full 65 questions: **43, 0, 22** in 2022; **31, 6, 28** in 2023; **35, 7, 23** in 2024; **42, 4, 19** in 2025; **47, 5, 13** in 2026. Every MSQ and every NAT was in a technical section. Across the five papers, **105 of the 275 technical questions** were NAT and **22** were MSQ.

FACT. Five MSQs had **exactly one correct option** in the official key: 2023 Q27, 2024 Q30, 2024 Q43, 2024 Q44 and 2026 Q31.

FACT. MSQ carries no negative marking and no partial credit. You score only by selecting every correct option and no incorrect one.

ANALYSIS. Nearly two technical questions in five have been numerical answers, with no options to eliminate and no negative marking. MSQ has been present in every paper since 2023 but never in large numbers. Prepare for between roughly 13 and 28 NAT questions and a handful of MSQs, and practise judging each MSQ option as a separate true-or-false statement — including the possibility that only one is true.

What this book will not do

It will not tell you it knows the paper. Three commitments follow from that, and they are kept everywhere in this book:

- **Facts, analysis and predictions are labelled separately**, as above.
- **Confidence is reported in four dimensions.** Being confident that economic load dispatch will be examined is not the same as being confident about a particular dispatch question. Every question in the bank carries Topic, Concept, Question-pattern and Exact-question confidence, and the last is capped, because GATE repeats concepts and not wording.
- **Absence of evidence is printed as absence of evidence.** Thirteen topic groups named in the 2027 syllabus produced no question in five papers. Each is said so plainly, with no history invented to fill the gap, and questions written on syllabus authority alone say that this is what they rest on.

It will not cite a question it has not counted. Every historical claim in this book names the year and the question number — "2026 (Q34)" — and Electrical Engineering sets one paper a year, so that citation identifies exactly one question in an official paper you can download.

It will not copy a question. Every question in this book was written for it, and every numerical answer was computed rather than estimated.

Section 7 — Control Systems

Mathematical modelling and representation of systems, Feedback principle, transfer function, Block diagrams and Signal flow graphs, Transient and Steady-state analysis of linear time invariant systems, Stability analysis using Routh-Hurwitz and Nyquist criteria, Bode plots, Root loci, Lag, Lead and Lead-Lag compensators; P, PI and PID controllers; State space model, Solution of state equations of LTI systems.

Section 8 — Electrical and Electronic Measurements

Bridges and Potentiometers, Measurement of voltage, current, power, energy and power factor; Instrument transformers, Digital voltmeters and multi-meters, Phase, Time and Frequency measurement; Oscilloscopes, Error analysis.

Section 9 — Analog and Digital Electronics

Simple diode circuits: clipping, clamping, rectifiers; Amplifiers: biasing, equivalent circuit and frequency response; oscillators and feedback amplifiers; operational amplifiers: characteristics and applications; single stage active filters, Active Filters: Sallen Key, Butterworth, VCOs and timers, combinatorial and sequential logic circuits, multiplexers, demultiplexers, Schmitt triggers, sample and hold circuits, A/D and D/A converters.

Section 10 — Power Electronics

Static V-I characteristics and firing/gating circuits for Thyristor, MOSFET, IGBT; DC to DC conversion: Buck, Boost and Buck-Boost Converters; Single and three-phase configuration of uncontrolled rectifiers; Voltage and Current commutated Thyristor based converters; Bidirectional ac to dc voltage source converters; Magnitude and Phase of line current harmonics for uncontrolled and thyristor based converters; Power factor and Distortion Factor of AC to DC converters; Single-phase and three-phase voltage and current source inverters, sinusoidal pulse width modulation.

***Note this, because it decides more marks than any typographical point above.** Several phrases appear in two sections. **Fourier series** is named in Section 1 (Calculus) and in Section 4; **divergence** in Section 1 (the divergence theorem) and in Section 3; **rectifiers** in Section 9 (simple diode circuits) and in Section 10 (uncontrolled rectifiers); **transient and steady-state analysis** in Section 2 (of networks) and in Section 7 (of linear time invariant systems). The syllabus does not say which section a question on such a phrase belongs to. Section 2.4 says how this book decides, once, for every paper.*

2.2 What changed between 2026 and 2027

FACT. The comparison below was made by a program that reads the two official PDFs and compares them phrase by phrase and word by word. No coaching site's "GATE 2027 syllabus changes" article was opened, quoted or checked against: a change list copied from another change list is worth nothing, and a wrong one is worse than none.

The method. Each section's text is split into phrases on its own commas, semicolons, colons and full stops. The 2026 and 2027 phrase lists are aligned after folding only case and spacing, so that a change of a single word or letter would appear as a reworded phrase — unless the only difference is a spelling correction, which is reported separately. Separately, each section is compared token by token — every word and every punctuation mark — which is what finds a changed capital or a dropped full stop that a phrase split cannot see. Finally the whole of each document is checked for any word that appears in one and nowhere in the other.

The result.

Section	2026 phrases	2027 phrases	Added	Removed	Reworded	Spelling corrected
1. Engineering Mathematics	53	53	0	0	0	0
2. Electric circuits	22	22	0	0	0	0
3. Electromagnetic Fields	20	20	0	0	0	0
4. Signals and Systems	12	12	0	0	0	0
5. Electrical Machines	31	31	0	0	0	0
6. Power Systems	20	20	0	0	0	0
7. Control Systems	14	14	0	0	0	0
8. Electrical and Electronic Measurements	11	11	0	0	0	0
9. Analog and Digital Electronics	21	21	0	0	0	1
10. Power Electronics	13	13	0	0	0	0
Total	217	217	0	0	0	1

Whole-document vocabulary check. Words in the 2026 syllabus that appear nowhere in the 2027 syllabus: "butterwoth". Words in the 2027 syllabus that appear nowhere in the 2026 syllabus: "butterworth". Nothing else.

Nine of the ten sections are identical word for word and punctuation mark for punctuation mark. The one entry in the last column is in Section 9:

	2026	2027
Section 9, active filters	"Active Filters: Sallen Key, Butterwoth"	"Active Filters: Sallen Key, Butterworth"

The layout changed in two further ways, neither touching what is named. The page header now reads "GATE 2027" and "IIT Madras | Organizing Institute", where 2026's read "GATE 2026 IIT Guwahati | Organizing Institute". And 2027 sets each section's name on its own line with a blank line between sections; both documents number the ten sections "Section 1:" to "Section 10:" with the same names. This book's section numbers are the 2027 document's own.

The two documents, for the record.

	2026	2027
Published by	IIT Guwahati	IIT Madras
Pages	2	2
Creation date in the PDF's metadata	2025-08-07	2026-07-06
Phrases	217	217
Tokens (words and punctuation marks)	923	923

2.3 What an unchanged syllabus means for how you study

(a) Three of the five counted papers were set on the same named topics as 2027

FACT. The official 2024 (IISc Bengaluru) and 2025 (IIT Roorkee) EE syllabi were compared with 2027's in the same way, phrase by phrase. Neither adds, removes or rewords a phrase. The 2025 text differs from 2027's only in the spelling "Butterwoth". The 2024 text differs in that spelling and in fourteen changes of capitalisation or punctuation

Section 3 — The Organising Institute, and the Five Papers It Did Not Set

GATE 2027 is organised by IIT Madras.

Each year one institute takes responsibility for the examination, and the paper is set under its supervision. The five Electrical Engineering papers this book counts were set under five different institutes, and **none of them was IIT Madras**. There is no paper in this book's record whose setter is the setter of the paper you are preparing for.

This chapter says so, says what that rules out, and then says what the five papers set under other institutes do show — the features of the paper that survived four changes of organiser. It is a short chapter because the honest version of it is short. A chapter that found an "IIT Madras style" in papers IIT Madras did not set would be inventing it.

3.1 What is fact here, and what is not

FACT. GATE 2027 is organised by IIT Madras. The official 2027 EE syllabus document carries the header "IIT Madras | Organizing Institute", and the official portal is gate2027.iitm.ac.in.

FACT. The organising institutes of the five counted EE papers are: **IIT Kharagpur (2022), IIT Kanpur (2023), IISc Bengaluru (2024), IIT Roorkee (2025) and IIT Guwahati (2026).**

FACT. IIT Madras last organised GATE in 2019, three years before the first paper this book counts.

NOT A FACT, and this book will not pretend otherwise. That an institute "has a style" which carries over into a paper it did not set, or across a gap of years and a change of everyone involved. Papers are set by subject panels, not by buildings. The 2019 Electrical Engineering paper set under IIT Madras is not part of this book's count, and this book does not guess at what it would show.

3.2 Five papers, five institutes

FACT. Technical marks by section, each paper counted question by question against its official key:

Section	2022, IIT Kharagpur	2023, IIT Kanpur	2024, IISc Bengaluru	2025, IIT Roorkee	2026, IIT Guwahati	Mean
1. Engineering Mathematics	15	13	13	15	18	14.8
2. Electric circuits	7	10	7	8	11	8.6
3. Electromagnetic Fields	5	4	4	3	6	4.4
4. Signals and Systems	9	11	11	9	4	8.8
5. Electrical Machines	10	8	6	9	9	8.4
6. Power Systems	10	10	12	9	12	10.6
7. Control Systems	8	9	9	9	6	8.2
8. Electrical and Electronic Measurements	1	2	3	5	2	2.6
9. Analog and Digital Electronics	9	10	9	8	7	8.6

Section	2022, IIT Kharagpur	2023, IIT Kanpur	2024, IISc Bengaluru	2025, IIT Roorkee	2026, IIT Guwahati	Mean
10. Power Electronics	11	6	11	10	10	9.6
Outside the 2027 syllabus	0	2	0	0	0	0.4
General Aptitude	15	15	15	15	15	15.0

ANALYSIS. In this record the organiser and the year are the same variable. Each institute set exactly one of the five papers, in exactly one year, so there is no way to tell whether 2023's 6 marks of Power Electronics reflect IIT Kanpur's panel, that year's draw, or neither. The same is true of 2026's 4 marks of Signals and Systems and 18 of Engineering Mathematics. **Any claim about what a particular institute "does" in EE rests, in this book's record, on one paper and cannot be separated from chance.** This book makes none.

3.3 What IIT Madras has already decided about EE

FACT. The one EE document IIT Madras has published so far is the 2027 syllabus. Its PDF metadata gives a creation date of 6 July 2026. Compared word by word with the 2026 document published by IIT Guwahati, it adds no phrase, removes no phrase and rewords no phrase. What it changes is one spelling — "Butterwoth" becomes "Butterworth" in Section 9 — and presentation: it sets each section's name on its own line and replaces the page header with its own (Section 2.2).

ANALYSIS. That is the only public decision about the EE paper that can be attributed to the 2027 organiser, and it was a decision to leave the examinable content exactly as it was. It supports planning on the 2026 syllabus and the five counted papers without adjustment. It supports nothing more specific than that.

3.4 What survived four changes of organiser

If no single institute's paper can tell you about IIT Madras, five institutes' papers can at least tell you what did not change when the institute did.

FACT. In all five papers, set under five different institutes:

Feature	2022	2023	2024	2025	2026
Questions / marks	65 / 100	65 / 100	65 / 100	65 / 100	65 / 100
One-mark / two-mark questions	30 / 35	30 / 35	30 / 35	30 / 35	30 / 35
General Aptitude	10 questions, 15 marks, all MCQ	the same	the same	the same	the same
Block layout	Q1–5 and Q11–35 one mark; Q6–10 and Q36–65 two marks	the same	the same	the same	the same
MCQ / MSQ / NAT	43 / 0 / 22	31 / 6 / 28	35 / 7 / 23	42 / 4 / 19	47 / 5 / 13
Section with most technical marks	Engineering Mathematics (15)	Engineering Mathematics (13)	Engineering Mathematics (13)	Engineering Mathematics (15)	Engineering Mathematics (18)
Every technical section examined	yes	yes	yes	yes	yes

Section 4 — What Five Papers Actually Contain

This is the chapter the rest of the book rests on. Everything after it — the subject chapters, the question bank, the mock tests — is built on the counts below, so this chapter shows you the counts, the method that produced them, and the places where the method needed a judgement.

It is also the chapter that has to say most plainly how little five papers can prove. ****The record is the five most recent GATE Electrical Engineering papers, 2022 to 2026, one a year.**** Five papers show patterns — what has been asked, how often, in what form. They do not show trends, and this chapter does not claim any.

4.1 How the counting was done

FACT. Five official GATE Electrical Engineering papers were read one question at a time: 2022, 2023, 2024, 2025 and 2026, **325 questions and 500 marks.**

Each question was read from the organising institute's own question paper — not from a key, not from a summary, not from a solved-papers book. Its marks, its question type and its answer were then taken from **that year's official answer key** and from nothing else. So when this book says a question carried two marks and was a numerical-answer question, that is the organisers' record of it and not a reading of the paper's layout.

Each question was then assigned to a section of the **2027** syllabus, and to a topic inside that section, in the syllabus's own words wherever the syllabus has words for it. That is the point of the exercise: a 2027 candidate needs to know where the marks would land under the syllabus they will actually sit.

Every count had to close, or it was not a count. Each of the five closes on:

Check	Required	Result
Questions	65	65 in all five
Total marks	100	100 in all five
One-mark questions	30	30 in all five
Two-mark questions	35	35 in all five
General Aptitude	exactly 15	15 in all five (10 questions)
Technical	exactly 85	85 in all five
Marks and type of every row	as in the official key	match in all five

A count that did not close would have been re-done rather than adjusted. Nothing was moved, merged or dropped to make a total come out right.

All five papers use the same layout, and the count confirms it question by question: Q1 to Q10 are General Aptitude, Q1 to Q5 at one mark and Q6 to Q10 at two; the technical questions run from Q11 to Q65, Q11 to Q35 at one mark and Q36 to Q65 at two. A citation such as "2025 (Q35)" in this book is the paper's own printed number.

Where a question could honestly sit in two sections, both are recorded. The question is placed by the boundary rules stated in Section 2.4, the other candidate section is recorded beside it, and where no rule settles the matter the question is marked as counted at its lean, with the reason written out. ****42 of the 275 technical questions**** carry an alternative, and thirty questions are counted at a lean. Section 4.7 prints the thirty and shows what moving them would do.

Why not use a published weightage table. Because a table that does not print its boundary decisions cannot be checked, and in EE the boundaries decide real marks: 67 of the 425 technical marks set in five papers sit on questions that could have been counted in a second section. A count that names every question it placed and where else it could have gone is a different kind of object from a table of totals.

4.2 The papers

FACT.

Year	Organising institute	Session recorded in the key	Question paper	Answer key
2026	IIT Guwahati	Session 5	counted	official key
2025	IIT Roorkee	Session 4	counted	official key
2024	IISc Bengaluru	Session 8	counted	official final key
2023	IIT Kanpur	Session 3	counted	official key
2022	IIT Kharagpur	Session 2	counted	official key, also printed at the end of the paper

Three facts about this base that you should have rather than discover:

- **The record starts in 2022.** Earlier Electrical Engineering papers exist and are not counted here. A "previous-year question" dated before 2022 is outside this book's evidence, and this book cites none.
- **Each year has one EE paper.** The key records a single session against every EE row, so a year and a question number identify a question completely.
- **Four questions were awarded to everyone.** The official keys mark 2023 (Q43), 2024 (Q7), 2024 (Q18) and 2026 (Q58) as MTA, marks to all. Each is counted in the section it was set in, because the paper awarded its marks.

4.3 Marks by section, every paper

FACT. Technical marks by section, in descending order of the five-paper mean. General Aptitude is 15 in every paper and is listed last.

Section	2022	2023	2024	2025	2026	Mean	Range	Share of 425 technical marks
1. Engineering Mathematics	15	13	13	15	18	14.8	13-18	17.4%
6. Power Systems	10	10	12	9	12	10.6	9-12	12.5%
10. Power Electronics	11	6	11	10	10	9.6	6-11	11.3%
4. Signals and Systems	9	11	11	9	4	8.8	4-11	10.4%
2. Electric circuits	7	10	7	8	11	8.6	7-11	10.1%
9. Analog and Digital Electronics	9	10	9	8	7	8.6	7-10	10.1%
5. Electrical Machines	10	8	6	9	9	8.4	6-10	9.9%
7. Control Systems	8	9	9	9	6	8.2	6-9	9.6%
3. Electromagnetic Fields	5	4	4	3	6	4.4	3-6	5.2%
8. Electrical and Electronic Measurements	1	2	3	5	2	2.6	1-5	3.1%
Outside the 2027 syllabus	0	2	0	0	0	0.4	0-2	0.5%
General Aptitude	15	15	15	15	15	15.0	15-15	—

FACT. Questions and question types by section, all five papers together:

Section	Questions	One-mark	Two-mark	MCQ	MSQ	NAT
1. Engineering Mathematics	47	20	27	21	10	16
2. Electric circuits	29	15	14	13	1	15
3. Electromagnetic Fields	13	4	9	8	1	4

How to work them

Attempt the question before you read anything below it. The solution is written to be read second. Read it first and you will recognise the method without being able to produce it — a gap a timed paper exposes and a read-through does not.

Work every NAT question to a number before you look at the band. Nearly two technical questions in five in the counted papers were numerical answers, with no options to narrow the field. The band tells you how precise you had to be; it is not a hint.

Treat each multiple-select question as four separate statements. Between 0 and 7 of the 65 questions in each counted paper were MSQs, and five of them had only one correct option. Decide each option on its own, true or false, before you look at how many you have ticked.

Treat the "Why This Question Is High Probability" line as the study instruction. In the author's judgement this is the most useful habit the book can recommend, and it is worth being blunt about why. The question is one instance of a concept; the line tells you which concept, on what evidence, and how stable its question shape has been. A candidate who works the question and ignores that line has practised one item. A candidate who reads it has been told what to go and learn.

Use the four confidence figures to allocate time, never to predict the paper. They are the author's estimates, not measurements, they rest on five papers, and Section 7 says so at length.

5.1 Engineering Mathematics

Engineering Mathematics is the largest technical section of the EE paper on the counted record. Across the five papers from 2022 to 2026 it drew **74 of the 425 technical marks — 17.4 per cent**, a mean of **14.8 marks** a paper, ranging from 13 (2023 and 2024) to 18 (2026). No other technical section averages more than 10.6.

The evidence is **47 questions and 74 marks**, counted question by question from the five official papers and checked against the official answer keys. EE sets one paper a year, so every citation below names the year and the printed question number — "2025 (Q53)" — and nothing else is needed to find it.

The official 2027 syllabus for this section

FACT — GATE 2027 EE, Section 1: Engineering Mathematics (official syllabus PDF, IIT Madras)

Linear Algebra: Matrix Algebra, Systems of linear equations, Eigen values, Eigen vectors.

Calculus: Mean value theorems, Theorems of integral calculus, Evaluation of definite and improper integrals, Partial Derivatives, Maxima and minima, Multiple integrals, Fourier series, Vector identities, Directional derivatives, Line integral, Surface integral, Volume integral, Stokes's theorem, Gauss's theorem, Divergence theorem, Green's theorem.

Differential Equations: First order equations (linear and nonlinear), Higher order linear differential equations with constant coefficients, Method of variation of parameters, Cauchy's equation, Euler's equation, Initial and boundary value problems, Partial Differential Equations, Method of separation of variables.

Complex Variables: Analytic functions, Cauchy's integral theorem, Cauchy's integral formula, Taylor series, Laurent series, Residue theorem, Solution integrals.

Probability and Statistics: Sampling theorems, Conditional probability, Mean, Median, Mode, Standard Deviation, Random variables, Discrete and Continuous distributions, Poisson distribution, Normal distribution, Binomial distribution, Correlation analysis, Regression analysis.

Two notes on that text. It is reproduced as printed, including "Eigen values" as two words and the capitalised "Partial Derivatives". "Sampling theorems" sits under **Probability and Statistics**, so here it means sampling in the statistical sense, not the signal-sampling theorem of Section 4.

5.1.1 What the 2027 syllabus changed here

FACT. The generated comparison of the official PDFs (`research/syllabus_changes.md`) reports for 2026 → 2027: *"Section 1 — Engineering Mathematics. No change. The section's text is identical word for word and punctuation mark for punctuation mark (after joining lines)."*

FACT. Against 2024 the only differences it finds in this section are two capitals: *"equations → Equations"* and *"variables → Variables"*. Against 2025 the section is identical token for token. The official 2022 and 2023 EE syllabi could not be obtained, so nothing is said about them.

ANALYSIS. For this section the 2027 syllabus is the syllabus that the 2024, 2025 and 2026 papers were set on, word for word. Every counted question below is therefore direct evidence, and no topic here is newly named: the items no paper has yet examined have been on the syllabus all along.

5.1.2 What the five papers actually asked

FACT. Every Section 1 row in the five count files.

Year	Questions	Marks	1-mark	2-mark	MCQ	MSQ	NAT
2022	8	15	1	7	7	0	1
2023	9	13	5	4	3	1	5
2024	9	13	5	4	2	5	2
2025	10	15	5	5	6	1	3
2026	11	18	4	7	3	3	5
Total	47	74	20	27	21	10	16

FACT. The same 74 marks by the syllabus's five paragraphs:

Topic	Questions	Marks	Share	2022	2023	2024	2025	2026	Papers
Calculus	15	28	38%	8	6	6	5	3	all five
Linear Algebra	15	20	27%	5	2	2	3	8	all five
Probability and Statistics	8	12	16%	2	4	1	3	2	all five
Differential Equations	5	8	11%	0	1	2	2	3	four (2023–2026)
Complex Variables	4	6	8%	0	0	2	2	2	three (2024–2026)

ANALYSIS — five papers show patterns, not trends. Three paragraphs — Calculus, Linear Algebra and Probability and Statistics — appeared in all five papers. Differential Equations appeared in four (absent in 2022) and Complex Variables in three (absent in 2022 and 2023). The 2026 total of 18 is the highest of the five and the 2026 Linear Algebra figure of 8 marks is more than any earlier year gave it, but one high year out of five is a swing, not a direction: 2023 and 2024 gave the section 13 marks each.

ANALYSIS — the type mix has no fixed habit. 2022 set seven of its eight questions as MCQ; 2024 set five of its nine as MSQ; 2023 and 2026 leaned on NAT. Over the five papers the split is 21 MCQ, 10 MSQ and 16 NAT. **MSQ is heavier here than in the paper as a whole** — 10 of 47 questions, against 22 of 275 technical questions overall — because statement-checking ("which properties hold?") suits pure mathematics.

ANALYSIS — the boundary. Eight counted questions sit on a boundary and were recorded as straddles. Counted **here** with another section as runner-up: 2022 (Q52), 2025 (Q51) and 2026 (Q32), pure divergence-theorem and vector-identity questions (runner-up Section 3); 2025 (Q54), a coupled pair of first-order equations (runner-up Section 7); and 2026 (Q14), classifying a differential equation (runner-up Section 4). Counted in **Section 4** with this section as runner-up: 2022 (Q46), 2024 (Q33) and 2026 (Q36), where the

Question Bank — Engineering Mathematics

Thirty-two questions, written for this book, in rough proportion to the counted marks: twelve on Calculus, eight on Linear Algebra, five on Probability and Statistics, four on Differential Equations and three on Complex Variables. Every numerical answer was computed. A matrix is written row by row, rows separated by semicolons: $[3 \ 1; 2 \ 2]$ has first row $(3, 1)$ and second row $(2, 2)$.

QEM.1

MCQ · 1 mark

Easy

Subject: Engineering Mathematics | Topic: Linear Algebra | Subtopic: Cayley–Hamilton theorem — inverse from the characteristic polynomial

Prediction Confidence: Topic 90% | Concept 74% | Question-pattern 62% | Exact-question 14%

Based on PYQs: GATE 2025 (Q14), 2022 (Q42) · last tested 2025 · new question on the same concept, not a reprint

QUESTION

For the matrix $A = [3 \ 1; 2 \ 2]$, with I the 2×2 identity matrix, A^{-1} equals

- (A) $(A - 5I)/4$
- (B) $-(A + 5I)/4$
- (C) $(5I - A)/4$
- (D) $(5I - A)/5$

ANSWER (C)

Concept Tested: Every square matrix satisfies its own characteristic equation, and multiplying that equation by A^{-1} gives the inverse as a polynomial in A .

Why This Question Is High Probability: Cayley–Hamilton was the tool in 2022 (Q42), which asked for A^{-1} in terms of A^2 and A , and in 2025 (Q14), which asked for P^2 in terms of P and I . Linear Algebra appeared in all five papers.

SOLUTION

Step 1 — characteristic polynomial. For a 2×2 matrix it is

$$\lambda^2 - (\text{trace})\lambda + \det:$$

$$\text{trace} = 3 + 2 = 5, \det = 3 \times 2 - 1 \times 2 = 4, \text{ so } \lambda^2 - 5\lambda + 4 = 0.$$

Step 2 — Cayley–Hamilton. $A^2 - 5A + 4I = 0$.

Step 3 — multiply by A^{-1} . $A - 5I + 4A^{-1} = 0$, so

$$A^{-1} = (5I - A)/4 = [2/4 \ -1/4; -2/4 \ 3/4] = [0.5 \ -0.25; -0.5 \ 0.75].$$

$$\text{Check: } A \times A^{-1} = [1.5 - 0.5, -0.75 + 0.75; 1 - 1, -0.5 + 1.5] = I. \text{ Answer (C).}$$

The wrong options, one mistake each.

- (A) drops the minus sign when $4A^{-1}$ is moved across: it is exactly $-A^{-1}$.
- (B) writes the characteristic polynomial as $\lambda^2 + 5\lambda + 4$, putting the trace in with the wrong sign.
- (D) divides by the trace, 5, instead of by the constant term, which is the determinant 4.

QEM.2

NAT · 1 mark

Easy

Subject: Engineering Mathematics | Topic: Linear Algebra | Subtopic: Sum of the eigenvalues of A^2

Prediction Confidence: Topic 90% | Concept 76% | Question-pattern 64% | Exact-question 15%

Based on PYQs: GATE 2024 (Q32), 2022 (Q40) · last tested 2024 · new question on the same concept, not a reprint

QUESTION

The matrix $A = [1 \ 2 \ 0; 3 \ 2 \ 1; 0 \ 4 \ 1]$. The sum of the eigenvalues of A^2 is ____ (in integer).

ANSWER 26**Acceptance band:** 26 to 26.**Concept Tested:** The eigenvalues of A^2 are λ^2 , and their sum is $\text{trace}(A^2)$ — found without computing a single eigenvalue.**Why This Question Is High Probability:** 2024 (Q32) asked the same quantity for a 2×2 matrix, and 2022 (Q40) asked for the eigenvalues of a function of a matrix (e^A). "Eigen values" is named in the syllabus.**SOLUTION**If $Ax = \lambda x$ then $A^2x = \lambda^2x$, so the eigenvalues of A^2 are $\lambda_1^2, \lambda_2^2, \lambda_3^2$, and their sum is the trace of A^2 :

$$\sum \lambda_i^2 = \text{trace}(A^2) = \sum_i \sum_j a_{ij} a_{ji}.$$

Step 1 — diagonal terms. $a_{11}^2 + a_{22}^2 + a_{33}^2 = 1 + 4 + 1 = 6$.**Step 2 — off-diagonal pairs**, each counted twice: $2(a_{12}a_{21} + a_{13}a_{31} + a_{23}a_{32}) = 2(2 \times 3 + 0 \times 0 + 1 \times 4) = 2 \times 10 = 20$.**Step 3 — total.** $\text{trace}(A^2) = 6 + 20 = 26$.**Check by the eigenvalues themselves:** A has eigenvalues 4.7016, 1 and -1.7016 , whose squares 22.105, 1 and 2.895 add to 26.**Two traps.** $(\text{trace } A)^2 = 4^2 = 16$ is the square of the sum, not the sum of the squares. And the sum of the squares of all nine entries, 36, is $\text{trace}(A^T A)$, which equals $\text{trace}(A^2)$ only when A is symmetric.**QEM.3**

MCQ · 2 marks

Moderate

Subject: Engineering Mathematics | Topic: Linear Algebra | Subtopic: Systems of linear equations — consistency and the number of solutions

Prediction Confidence: Topic 90% | Concept 80% | Question-pattern 66% | Exact-question 15%

Based on PYQs: GATE 2026 (Q56), 2025 (Q13) · last tested 2026 · new question on the same concept, not a reprint

QUESTIONThe system of equations in the real unknowns x, y, z

$$x + y + z = 3$$

$$x + 2y + 3z = 6$$

$$x + 4y + kz = \mu$$

has infinitely many solutions if and only if

(A) $k = 7$ and $\mu \neq 12$

(B) $k = 7$ and $\mu = 12$

(C) $k \neq 7$, for every μ

(D) $k = 7$, for every μ

ANSWER (B)**Concept Tested:** The rank test: unique solution when $\text{rank}[A] = 3$; infinitely many when $\text{rank}[A] = \text{rank}[A | b] < 3$; none when $\text{rank}[A] < \text{rank}[A | b]$.**Why This Question Is High Probability:** 2025 (Q13) asked the number of solutions of $Ax = b$ and 2026 (Q56) the consequences of a unique solution. "Systems of linear equations" is named.**SOLUTION**Row-reduce the augmented matrix $[A | b] = [1 \ 1 \ 1 | 3; 1 \ 2 \ 3 | 6; 1 \ 4 \ k | \mu]$.**Step 1.** $R_2 - R_1$ and $R_3 - R_1$: $[1 \ 1 \ 1 | 3; 0 \ 1 \ 2 | 3; 0 \ 3 \ k-1 | \mu-3]$.**Step 2.** $R_3 - 3R_2$: $[1 \ 1 \ 1 | 3; 0 \ 1 \ 2 | 3; 0 \ 0 \ k-7 | \mu-12]$.**Step 3 — read the last row**, $(k - 7)z = \mu - 12$:

Case	rank[A]	rank[A b]	Solutions
$k \neq 7$	3	3	unique
$k = 7, \mu = 12$	2	2	infinitely many (one free unknown)
$k = 7, \mu \neq 12$	2	3	none

Answer **(B)**.

The wrong options, one mistake each.

- **(A)** is the no-solution case: it confuses a contradictory last row, $0 = \mu - 12 \neq 0$, with a free one.
- **(C)** is the unique-solution case: a non-zero pivot in every row leaves nothing free.
- **(D)** stops at $\det A = 0$ (which happens at $k = 7$) and forgets the consistency check on μ . A singular coefficient matrix alone never decides between none and infinitely many.

QEM.4

MSQ · 2 marks

Moderate

Subject: Engineering Mathematics | Topic: Linear Algebra | Subtopic: Eigenvalues and eigenvectors of a real symmetric matrix

Prediction Confidence: Topic 90% | Concept 78% | Question-pattern 62% | Exact-question 13%

Based on PYQs: GATE 2026 (Q30), 2025 (Q12), 2024 (Q32) · last tested 2026 · new question on the same concept, not a reprint

QUESTION

A is a 3×3 real symmetric matrix with eigenvalues 1, 2 and 4, and I is the 3×3 identity matrix. Which of the following statement(s) is/are TRUE?

- (A) $A^2 - 5A + 4I$ is singular
 (B) Eigenvectors of A belonging to the eigenvalues 1 and 4 are orthogonal (C) A^{-1} exists and its eigenvalues are 1, 0.5 and 0.25 (D) The sum of the eigenvalues of A^2 is 49

ANSWER (A), (B), (C)

Concept Tested: The spectral mapping $p(A) \leftrightarrow p(\lambda)$, orthogonality of eigenvectors of a symmetric matrix, and the eigenvalues of A^{-1} .

Why This Question Is High Probability: 2025 (Q12) tested orthogonal eigenvectors of a real symmetric matrix, 2026 (Q30) the consequences of a shared eigenvalue, and 2024 (Q32) the eigenvalues of A^2 . Linear Algebra has had an MSQ in 2026 (Q30), (Q55) and (Q56).

SOLUTION

(A) TRUE. The eigenvalues of $p(A) = A^2 - 5A + 4I$ are $p(\lambda) = \lambda^2 - 5\lambda + 4 = (\lambda - 1)(\lambda - 4)$. So $p(1) = 0$, $p(2) = -2$, $p(4) = 0$, and $\det p(A) = 0 \times (-2) \times 0 = 0$. The matrix is singular.

(B) TRUE. For a real symmetric matrix, eigenvectors of distinct eigenvalues are orthogonal: from $Ax_1 = \lambda_1 x_1$ and $Ax_2 = \lambda_2 x_2$, $x_2^T Ax_1 = \lambda_1 x_2^T x_1$ and also $= \lambda_2 x_2^T x_1$, so $(\lambda_1 - \lambda_2)x_2^T x_1 = 0$ and, since $\lambda_1 \neq \lambda_2$, $x_2^T x_1 = 0$.

(C) TRUE. $\det A = 1 \times 2 \times 4 = 8 \neq 0$, so A^{-1} exists, with the same eigenvectors and eigenvalues $1/\lambda$: 1, 0.5 and 0.25.

(D) FALSE. The eigenvalues of A^2 are 1, 4 and 16, which sum to **21**. The value 49 is $(1 + 2 + 4)^2$, the square of the trace rather than the trace of the square.

Section 9 — The Hundred Questions to Work First

If you have time for a hundred questions and not for 264, these are the hundred. If you have time for all 264, work these first anyway, because finishing them leaves you in a defensible position if the time runs out.

9.1 How the hundred were chosen

This list is not hand-picked. One rule was applied to all 264 questions in the eleven subject chapters and the result printed. The rule is stated here so that you can disagree with it precisely.

How many come from each section. Each section's mean marks per paper across the five counted papers, rounded so that the quotas sum to exactly 100:

Section	Mean marks per paper	Questions in the hundred
General Aptitude	15.0	15
Engineering Mathematics	14.8	15
Power Systems	10.6	11
Power Electronics	9.6	10
Signals and Systems	8.8	9
Electric circuits	8.6	9
Analog and Digital Electronics	8.6	8
Electrical Machines	8.4	8
Control Systems	8.2	8
Electromagnetic Fields	4.4	4
Electrical and Electronic Measurements	2.6	3

So the hundred mirror the paper. An hour spent on this list is divided the way the marks have been. The means rest on five papers, and a section's marks moved by several between papers, so read each quota as a reasonable share rather than a forecast.

Which questions within a section. Topic confidence highest first — of the four confidence figures this is the one with the most evidence behind it, and Section 7 explains what it is and is not. Ties break on Question-pattern confidence, because a stable question *shape* is a skill that practice transfers to. Any remaining ties break on question number, so the order does not shuffle between printings.

Nothing else enters. Difficulty does not, because an easy question on a near-certain topic is worth more than a hard one on an unlikely topic, and this list is about return on an hour rather than about challenge.

Why the quota is there. Ranked on confidence alone, with no quota, the same rule would have taken 28 from General Aptitude, 25 from Engineering Mathematics, 9 from Power Electronics, 8 from Electrical Machines, 8 from Signals and Systems, 7 from Power Systems, 6 from Analog and Digital Electronics, 5 from Electric circuits and 4 from Control Systems. That would have short-changed Electric circuits (5 against a quota of 9), Electromagnetic Fields (0 against a quota of 4), Power Systems (7 against a quota of 11) and Control Systems (4 against a quota of 8) — not because those sections matter less, but because fewer of their questions carry the very highest Topic figures. A study list that under-weights a section the paper reliably examines is wrong however cleanly it was derived, so the quota decides how many and the confidence figures decide which.

9.2 The blind spot in the rule, stated before the table

The rule ranks on Topic confidence, and Topic confidence rests mostly on the counted record. A topic the syllabus names that none of the five papers has set — oscilloscopes (Electrical and Electronic Measurements), signal flow graphs (Control Systems) and Schmitt triggers (Analog and Digital Electronics) among them — has ****no counted history at all**, and so some of the lowest confidence figures in the book**, and its questions drop to the bottom of this list or off it.

That is backwards as a study instruction. A never-examined topic is exactly the material no amount of past-paper practice will make you meet, so it is the material most likely to be missing from your preparation on the day. Section 9.4 lists those questions separately, together with every other question whose concept the counted record has never examined. **Work the hundred, then work Section 9.4.** Do not let the ranking decide that a named topic is unimportant; it decided that only because five papers have not yet set it.

9.3 The hundred

Mk is the question's marks. **T** and **P** are its Topic and Question-pattern confidence. Work the question from the subject chapter, not from this table — the table is an index, and the chapters hold the solutions, the acceptance bands and the reasoning.

#	Question	Section	Topic — subtopic	Type	Mk	T	P
1	QGA.10	General Aptitude	Quantitative aptitude — Successive percentages and a delivery charge	MCQ	1	90%	58%
2	QGA.13	General Aptitude	Quantitative aptitude — Probability with two dice	MCQ	2	90%	58%
3	QGA.11	General Aptitude	Quantitative aptitude — Laws of exponents	MCQ	1	90%	56%
4	QGA.17	General Aptitude	Quantitative aptitude — Intersection of a line and a parabola	MCQ	1	90%	56%
5	QGA.8	General Aptitude	Quantitative aptitude — Ratio of speeds in a race	MCQ	1	90%	54%
6	QGA.12	General Aptitude	Quantitative aptitude — Unit digit of a product of powers	MCQ	2	90%	54%
7	QGA.9	General Aptitude	Quantitative aptitude — Number systems (decimal to hexadecimal)	MCQ	1	90%	48%
8	QGA.16	General Aptitude	Quantitative aptitude — Heights and distances	MCQ	2	90%	48%
9	QGA.15	General Aptitude	Quantitative aptitude — Sum of an arithmetic progression	NAT	1	90%	34%
10	QGA.14	General Aptitude	Quantitative aptitude — Permutations with a restriction	NAT	2	90%	32%
11	QGA.1	General Aptitude	Verbal aptitude — Word analogy (antonyms)	MCQ	1	88%	62%
12	QGA.7	General Aptitude	Verbal aptitude — Reading comprehension — valid inference	MCQ	2	88%	62%
13	QGA.2	General Aptitude	Verbal aptitude — Reported speech (backshift of tense and modal)	MCQ	1	88%	60%
14	QGA.4	General Aptitude	Verbal aptitude — Prepositions (fixed collocations)	MCQ	2	88%	56%
15	QGA.5	General Aptitude	Verbal aptitude — Matching words with synonyms	MCQ	2	88%	54%
16	QEM.12	Engineering Mathematics	Calculus — Directional derivative	MCQ	2	92%	74%
17	QEM.16	Engineering Mathematics	Calculus — Line integral by Green's theorem	NAT	2	92%	66%
18	QEM.17	Engineering Mathematics	Calculus — Divergence theorem	NAT	1	92%	66%
19	QEM.11	Engineering Mathematics	Calculus — Evaluation of an improper integral	NAT	2	92%	60%
20	QEM.13	Engineering Mathematics	Calculus — Maxima and minima of a function of two variables	MCQ	1	92%	60%
21	QEM.15	Engineering Mathematics	Calculus — Vector identities	MSQ	2	92%	60%
22	QEM.9	Engineering Mathematics	Calculus — Theorems of integral calculus — monotonicity of an integral-defined function	MCQ	2	92%	58%
23	QEM.20	Engineering Mathematics	Calculus — Continuity and differentiability of x	MSQ	1	92%	54%
24	QEM.19	Engineering Mathematics	Calculus — Fourier series of a function on $(-\pi, \pi)$	MCQ	1	92%	52%
25	QEM.14	Engineering Mathematics	Calculus — Multiple integrals — changing the order of integration	NAT	2	92%	50%
26	QEM.18	Engineering Mathematics	Calculus — Stokes's theorem	MCQ	2	92%	46%
27	QEM.10	Engineering Mathematics	Calculus — Lagrange's mean value theorem	MCQ	1	92%	40%
28	QEM.3	Engineering Mathematics	Linear Algebra — Systems of linear equations — consistency and the number of solutions	MCQ	2	90%	66%
29	QEM.2	Engineering Mathematics	Linear Algebra — Sum of the eigenvalues of A^2	NAT	1	90%	64%
30	QEM.7	Engineering Mathematics	Linear Algebra — Quadratic form of a skew-symmetric matrix	NAT	1	90%	64%
31	QPS.12	Power Systems	Economic Load Dispatch — Equal incremental cost with a generator limit	NAT	2	85%	66%
32	QPS.21	Power Systems	Symmetrical and unsymmetrical fault analysis — Single line-to-ground fault through a neutral reactor	NAT	2	85%	60%
33	QPS.19	Power Systems	Symmetrical components — Positive sequence component of unbalanced currents	NAT	1	85%	58%
34	QPS.22	Power Systems	Symmetrical and unsymmetrical fault analysis — Bus voltage during a symmetrical fault by Zbus	MCQ	2	85%	58%
35	QPS.13	Power Systems	Economic Load Dispatch — Effect of a fixed-cost change on dispatch	MCQ	1	85%	56%
36	QPS.20	Power Systems	Symmetrical components — Zero sequence paths through transformers	MCQ	1	85%	54%
37	QPS.14	Power Systems	Economic Load Dispatch — Dispatch with transmission losses and penalty factor	MCQ	2	85%	52%
38	QPS.9	Power Systems	Gauss-Seidel and Newton-Raphson load flow methods — Size of the Jacobian	MCQ	1	80%	60%
39	QPS.2	Power Systems	Models and performance of transmission lines — Nominal- π model, sending-end voltage	NAT	2	80%	58%
40	QPS.3	Power Systems	Models and performance of transmission lines — Real and reactive power flow on a lossless line	MSQ	2	80%	56%

Section 10 — What Not to Skip, and What to Skip

There are two ways to be wrong about a syllabus. You can study something it does not contain, which costs time. You can skip something it does contain, which costs marks. **These are not symmetric**, and treating them as though they were is the mistake behind most reading lists.

The asymmetry has a practical consequence that runs through this whole chapter: where a topic's status is genuinely unclear, read it briefly rather than skip it. An hour lost is recoverable. A question you cannot attempt is not.

This chapter gives two lists of what not to skip, built from opposite kinds of evidence, and then says what can be skipped. The last answer is shorter than in most books of this kind, for a reason Section 10.3 gives.

10.1 Do not skip, group one: the topics examined in all five counted papers

The strongest signal five papers can give is a topic that drew marks in **every one of them** — 2022, 2023, 2024, 2025 and 2026, five different setting institutes, no gap. Five papers show patterns, not trends, and this is the clearest pattern they show: a topic the examiners have not once left out.

14 of the 89 topic groups in the counted record did so. They carry 207 of the 500 marks counted across the five papers. If your preparation is incomplete, it should be incomplete somewhere other than here.

Topic	Section	Counted marks	Papers
Calculus: single-variable, multiple integrals, vector calculus and integral theorems	Engineering Mathematics	28	5 of 5
Quantitative aptitude	General Aptitude	23	5 of 5
Linear algebra: matrix algebra, linear equations, eigenvalues and eigenvectors	Engineering Mathematics	20	5 of 5
Verbal aptitude	General Aptitude	18	5 of 5
Three-phase induction machines	Electrical Machines	17	5 of 5
Analytical aptitude	General Aptitude	16	5 of 5
Spatial aptitude	General Aptitude	16	5 of 5
Operational amplifiers: characteristics and applications	Analog and Digital Electronics	14	5 of 5
Probability and statistics: probability, random variables, distributions, correlation, regression	Engineering Mathematics	12	5 of 5
Transient response of DC and AC networks	Electric circuits	11	5 of 5
DC to DC converters: buck, boost, buck-boost	Power Electronics	10	5 of 5
Economic load dispatch	Power Systems	8	5 of 5
Thyristor, MOSFET and IGBT: static V-I characteristics and gating	Power Electronics	7	5 of 5
Symmetrical components and fault analysis	Power Systems	7	5 of 5

A caution about the topics *not* in that table

****A low paper-count is much weaker evidence than a high one, and this book draws no skip conclusion from it.****
There are two reasons.

The first is the size of the record. With five papers, a topic set once is as consistent with *rare* as with *not drawn yet*, and nothing in the record can tell the two apart.

The second is mechanical. Every counted question is placed in exactly one section and one topic, and EE's sections overlap by design — Laplace transforms inside circuits and control, a machine inside a power network, a converter driving a motor — so a skill used inside questions counted elsewhere shows a low count of its own without being rare in the examination. The sharpest case in the record is Electromagnetic Fields in 2025: the section was counted at **3 marks**, while 4 more marks in the same paper — 2025 (Q51) and 2025 (Q52) — recorded it as the runner-up and were counted where their method sits. A reader of the section total alone would conclude that the paper hardly asked for it. It asked for it more than the total shows.

Read the table above as *these are certain*. Do not read its absences as *these are safe*.

10.2 Do not skip, group two: what the syllabus names and no paper has set

These have no past-paper history at all, so ****no amount of previous-year practice will make you meet them****. That is exactly why they belong on a do-not-skip list: they are the material most likely to be missing from a preparation built on past papers.

First, the whole topic groups that have drawn no marks in the five papers:

Section	Topic named in the syllabus, with no counted marks
Electromagnetic Fields	Inductance, magnetomotive force, reluctance and magnetic circuits
Signals and Systems	R.M.S. and average value of a periodic waveform
Power Systems	Power generation concepts, AC and DC transmission concepts
Power Systems	Series and shunt compensation
Power Systems	Electric field distribution and insulators
Power Systems	Per-unit quantities
Electrical and Electronic Measurements	Digital voltmeters and multimeters
Electrical and Electronic Measurements	Phase, time and frequency measurement
Electrical and Electronic Measurements	Oscilloscopes
Analog and Digital Electronics	Oscillators and feedback amplifiers
Analog and Digital Electronics	Active filters
Analog and Digital Electronics	VCOs and timers
Analog and Digital Electronics	Schmitt triggers, sample and hold circuits

Three of these rows are zeros of the counting rule rather than of the papers, because every counted question sits in one group only: an inductance was computed from a core's geometry and permeability in 2023 (Q53), counted in Electrical Machines; the R.M.S. value of a triangular current was asked in 2026 (Q47), counted in Power Electronics, and a mean-square value in 2025 (Q30), counted under Fourier series; per-unit values are the working units of the fault question 2025 (Q33). What no paper has set is a question on any of the three in its own right, so they stay on this list, but do not read those rows as untouched.

Second, the named items inside topic groups that have drawn marks, where no counted question's recorded topic names the item itself. These are finer-grained, and a question on a neighbouring item may have touched one in passing, so read this list as *not yet set as a question in its own right*:

Section	Named in the syllabus, not yet set on its own
Engineering Mathematics	Mean value theorems
Engineering Mathematics	Stokes's theorem
Engineering Mathematics	Method of variation of parameters

Section 11 — The Ten Mock Tests

Ten full papers follow. Each is 65 questions, 100 marks and three hours, built to the GATE Electrical Engineering pattern and to the section weights of the five counted papers. Every question in them was written for this book; none reproduces a past paper and none repeats a question from the subject chapters, which is why no mock question carries a year as a source.

A mock test is not a question bank with a clock attached. It measures three things the bank cannot: whether you recognise a topic when nothing announces it, whether you can hold accuracy for three hours, and where your time actually goes as opposed to where you think it goes. Use them for that.

The common blueprint

All ten papers are laid out identically, because the real paper is: the 2022, 2023, 2024, 2025 and 2026 papers share this layout exactly.

Questions	Block	Marks each
Q1 – Q5	General Aptitude	1
Q6 – Q10	General Aptitude	2
Q11 – Q35	Technical	1
Q36 – Q65	Technical	2

That gives 30 one-mark and 35 two-mark questions, General Aptitude at exactly 15 marks and the technical sections at 85 — the shape all five counted papers close on. In all five, every General Aptitude question was a multiple-choice question, and the mocks keep that too.

The 85 technical marks are distributed as the five counted papers distribute them. The target for each section is its five-paper mean, rounded to whole marks:

Section	2022	2023	2024	2025	2026	Mean	Target in each mock
Engineering Mathematics	15	13	13	15	18	14.8	15
Power Systems	10	10	12	9	12	10.6	11
Power Electronics	11	6	11	10	10	9.6	10
Signals and Systems	9	11	11	9	4	8.8	9
Electric circuits	7	10	7	8	11	8.6	9
Analog and Digital Electronics	9	10	9	8	7	8.6	8
Electrical Machines	10	8	6	9	9	8.4	8
Control Systems	8	9	9	9	6	8.2	8
Electromagnetic Fields	5	4	4	3	6	4.4	4
Electrical and Electronic Measurements	1	2	3	5	2	2.6	3

The ten means add to 84.6, not 85, because two marks of the 2023 paper — Q57, a microcontroller question — sit on material the 2027 syllabus does not name. The mocks set nothing outside the 2027 syllabus, so those two marks are given back to the named sections: the means are rounded by largest remainder so that the targets add to exactly 85. Electric circuits and Analog and Digital Electronics are tied exactly, at 43 marks each over the five papers; the tie is broken mechanically by syllabus section number, which gives the extra mark to Electric circuits (Section 2). Either section may carry 8 or 9 marks in a given mock.

Two consequences are worth saying out loud. ****The mocks vary the weights a little, but far less than the real paper has.**** A mock may move a section by a mark or two from its target, as a real paper does, and never by more than three. The five real papers moved much further: Signals and Systems ran from 11 marks in 2023 and 2024 to 4 in 2026, and Power Electronics from 11 in 2022 to 6 in 2023. A practice paper that swung that far would mix two measurements at once — your preparation and the paper's shape — so the mocks do not, but it means

no mock can show you how a four-mark Signals year feels. **And nothing in the mocks has to be set aside for syllabus changes**, because there were none that matter: the 2027 syllabus adds, removes and rewords no phrase against 2026's, and its only change of wording is one corrected spelling, "Butterwoth" to "Butterworth".

Marking, exactly as the official keys apply it.

Type	Correct	Wrong	Partly correct
MCQ	full marks	minus one-third of the question's marks	not possible
MSQ	full marks	no penalty	no credit
NAT	full marks for any value in the band	no penalty	not applicable

So a wrong one-mark MCQ costs $1/3$ and a wrong two-mark MCQ costs $2/3$. MSQ scores only when every correct option is selected and no incorrect one is — no partial credit, no negative marking — which makes it the format that punishes partial knowledge hardest. NAT carries no penalty at all, so a NAT left blank is strictly worse than a NAT answered.

Each paper states its own conventions — that AC voltages and currents are RMS unless stated, that three-phase voltages are line-to-line unless stated, that op-amps are ideal unless stated, that $j = \sqrt{-1}$, and that SI units are used throughout — and every numerical-answer question prints its acceptance band. The book has no drawings, so every circuit is given as a short netlist, one element per line with its nodes and node 0 as the reference, and every waveform or block diagram is described in words or in a small table.

Where a question depends on a sign convention, a reference direction or a base quantity, the stem states it, because without it the answer is not unique.

The papers are graded in difficulty: about a quarter of each paper easy, half moderate and a quarter hard, with at least two genuinely hard two-mark questions in every paper. That mix is the author's judgement, not a count; the official keys record no difficulty.

Why the MSQ and NAT counts vary between papers

The number of multiple-select and numerical-answer questions is different in each of the ten papers, and that is deliberate, because it is different in each real paper. Counted from the official answer keys:

Paper	MCQ	MSQ	NAT
2022	43	0	22
2023	31	6	28
2024	35	7	23
2025	42	4	19
2026	47	5	13

Because General Aptitude has been all MCQ, every MSQ and every NAT in those papers sat in the technical 85 marks. Across the five papers the range is 31 to 47 MCQ, 0 to 7 MSQ and 13 to 28 NAT. The mocks stay inside it and move within it from paper to paper: between about 35 and 43 MCQ, 4 and 7 MSQ and 18 and 24 NAT in each. They keep at least four MSQs, although 2022 set none, because every paper since then has set between four and seven. A set of mocks with constant counts would train you for a paper that does not exist.

Varying them means some of these ten papers will feel harsher than others for reasons that have nothing to do with your preparation — which is exactly what the record supports. A paper heavy in NAT is the one where elimination cannot help you: 28 of the 2023 paper's 55 technical questions had no options at all.

Mock Test 01

Duration: 180 minutes | Maximum marks: 100 | 65 questions

The layout of the real EE paper: Q1–Q5 General Aptitude at 1 mark, Q6–Q10 General Aptitude at 2 marks, Q11–Q35 technical at 1 mark, Q36–Q65 technical at 2 marks. Inside each technical block the multiple-choice questions come first, then the multiple-select, then the numerical-answer questions, as in the real paper.

This paper carries five multiple-select questions and twenty-two numerical-answer questions — the middle of the range seen across the five counted EE papers, where the MSQ count has run from none to seven and the NAT count from thirteen to twenty-eight. Its section marks sit exactly on the five-paper targets, so whatever it shows about a section is not an accident of this paper's weights.

Marking. MCQ: correct +marks, wrong $-1/3$ of marks. MSQ: all correct options and no incorrect one, or nothing; no negative marking. NAT: no negative marking; any value in the printed band scores.

Conventions. AC voltages and currents are RMS unless stated. Three-phase voltages are line-to-line unless stated. Op-amps and diodes are ideal unless stated. $j = \sqrt{-1}$. $u(t)$ and $u[n]$ are the unit steps, and $\exp(x)$ means e raised to the power x . In a netlist, node 0 is the reference. SI units throughout.

Q1 – Q5 carry ONE mark each.

Q1

MCQ · 1 mark

Select the option that completes the sentence correctly.

"Neither the supervisor nor the technicians _____ aware, until the alarm sounded, that the feeder had tripped."

- ☐ A was ☐ B were
☐ C is ☐ D has been

General Aptitude — verbal: agreement with the nearer subject

Q2

MCQ · 1 mark

Thermometer : temperature :: Anemometer : _____

- ☐ A rainfall ☐ B humidity
☐ C atmospheric pressure ☐ D wind speed

General Aptitude — verbal: instrument and quantity analogy

Q3

MCQ · 1 mark

The average of eleven numbers is 42. The average of the first six of them is 37 and the average of the last six is 46. The sixth number is

- ☐ A 36 ☐ B 41.5
☐ C 42 ☐ D 47

General Aptitude — quantitative: averages of overlapping groups

Q4

MCQ · 1 mark

The next term of the sequence 3, 7, 16, 35, 74, ... is

- ☐ A 113 ☐ B 148
☐ C 153 ☐ D 154

General Aptitude — analytical: number sequence

Q5 MCQ · 1 mark

An analogue wall clock is seen in a plane mirror hung on the opposite wall, and its image appears to show 3:40. The actual time is

- ☐ A 8:20 ☐ B 8:40
☐ C 9:20 ☐ D 9:40

General Aptitude — spatial: mirror image of a clock

Q6 – Q10 carry TWO marks each.

Q6 MCQ · 2 marks

The table gives the energy a distribution company sold to four categories of consumer in one year, and the tariff each category paid.

Category	Energy sold (GWh)	Tariff (rupees per kWh)
Domestic	120	4.50
Commercial	60	8.00
Industrial	200	6.50
Agricultural	80	2.00

The category whose share of the company's revenue exceeds its share of the energy sold by the largest number of percentage points is

- ☐ A Domestic ☐ B Commercial
☐ C Agricultural ☐ D Industrial

General Aptitude — data interpretation: shares from a table

Q7 MCQ · 2 marks

A committee of four is to be chosen from five engineers and four managers. The number of possible committees that contain at least two engineers is

- ☐ A 60 ☐ B 105
☐ C 125 ☐ D 126

General Aptitude — quantitative: combinations with a condition

Q8 MCQ · 2 marks

Read the passage and answer the question.

Preventive maintenance replaces a component after a fixed interval, whether or not it shows any wear. Condition-based maintenance replaces it only when measurements show that it is deteriorating. The second approach wastes less of a component's useful life, but it depends on sensors and on someone reading what they report; where the measurements cannot be relied upon, a fixed interval is the safer policy.

Q63 NAT · 2 marks

A Schering bridge ABCD is supplied at 50 Hz across A and C, with the detector between B and D. Arm AB is the capacitor under test, modelled as a capacitance in series with a resistance. Arm AD is a loss-free standard capacitor of 400 pF. Arm BC is a 250 Ω resistor. Arm DC is a 750 Ω resistor in parallel with a 0.2 μ F capacitor. At balance, the dissipation factor ($\tan \delta$) of the capacitor under test is ____ (round off to three decimal places).

Acceptance band: 0.046 to 0.048.

Electrical and Electronic Measurements — bridges: Schering bridge

Q64 NAT · 2 marks

An ideal op-amp has a 60 k Ω resistor connected from its output to its inverting input. A source of 1.2 V drives the inverting input through 12 k Ω . A source of 2.0 V drives the non-inverting input through 15 k Ω , and a 45 k Ω resistor joins the non-inverting input to ground. The output voltage is ____ V (round off to one decimal place).

Acceptance band: 2.9 to 3.1.

Analog and Digital Electronics — operational amplifiers: difference amplifier with unequal ratios

Q65 NAT · 2 marks

A single-phase full-bridge inverter on a 300 V DC link is phase-shift controlled so that, in each cycle, its output voltage is +300 V for 120°, zero for 60°, -300 V for 120° and zero for 60°. The total harmonic distortion of the output voltage is ____ per cent (round off to one decimal place).

Acceptance band: 30.6 to 31.6.

Power Electronics — single-phase voltage source inverter: quasi-square output

Answer key — Mock Test 01

Attempt the whole paper before reading this page. Score with the official scheme: a correct 1-mark question gives +1 and a correct 2-mark question +2; a wrong MCQ costs 1/3 of a mark (1-mark) or 2/3 of a mark (2-mark); MSQ and NAT carry no negative marking, and an MSQ scores only if every correct option is selected and no incorrect one is. NAT answers are given to the precision stated in each question.

This paper contains 38 MCQ, 5 MSQ and 22 NAT questions (30 one-mark and 35 two-mark, 100 marks).

Q	Answer	Type	Q	Answer	Type	Q	Answer	Type
1	B	MCQ	23	B	MCQ	45	B	MCQ
2	D	MCQ	24	A	MCQ	46	B	MCQ
3	A	MCQ	25	A,B,D	MSQ	47	D	MCQ
4	C	MCQ	26	A,C,D	MSQ	48	C	MCQ
5	A	MCQ	27	18	NAT	49	C	MCQ
6	D	MCQ	28	0.187	NAT	50	A,B,C	MSQ
7	B	MCQ	29	25.0	NAT	51	B,C,D	MSQ
8	A	MCQ	30	2.5	NAT	52	A,C	MSQ
9	D	MCQ	31	3.5	NAT	53	202	NAT
10	B	MCQ	32	216	NAT	54	0.407	NAT
11	D	MCQ	33	127.6	NAT	55	4.82	NAT
12	A	MCQ	34	0.30	NAT	56	9.21	NAT
13	D	MCQ	35	167	NAT	57	71.5	NAT
14	B	MCQ	36	D	MCQ	58	4.85	NAT
15	D	MCQ	37	C	MCQ	59	24.4	NAT
16	C	MCQ	38	B	MCQ	60	5.43	NAT
17	C	MCQ	39	A	MCQ	61	1.06	NAT
18	A	MCQ	40	A	MCQ	62	0.49	NAT
19	A	MCQ	41	D	MCQ	63	0.047	NAT
20	C	MCQ	42	C	MCQ	64	3.0	NAT
21	C	MCQ	43	B	MCQ	65	31.1	NAT

Section 13 — Detailed Mock Test Solutions

Each test's solutions follow, in question order. Attempt and score the paper (Section 12) before reading these.

Mock Test 01

Q1**ANSWER B**

Confidence: Pattern 48% | Exact-question 11%

With *neither ... nor*, the verb agrees with the subject nearer to it. That subject is *the technicians*, which is plural, and the rest of the sentence ("had tripped", "sounded") is in the past. So the verb is **were**.

A. was agrees with *the supervisor*, the farther subject. **C. is** is singular and also in the wrong tense. **D. has been** is singular, and the present perfect does not sit with "until the alarm sounded".

Q2**ANSWER D**

Confidence: Pattern 48% | Exact-question 11%

The relation is *instrument : the quantity it measures*. A thermometer measures temperature; an anemometer measures **wind speed**.

A. rainfall is measured by a rain gauge, **B. humidity** by a hygrometer and **C. atmospheric pressure** by a barometer. Each is a weather quantity, which is what makes it tempting, but each belongs to a different instrument.

Q3**ANSWER A**

Confidence: Pattern 48% | Exact-question 11%

The first six numbers add to $6 \times 37 = 222$ and the last six to $6 \times 46 = 276$. Together these two groups count every number once, except the sixth, which they count twice. All eleven add to $11 \times 42 = 462$. So

sixth number = $222 + 276 - 462 = 36$.

B. 41.5 is the mean of the two group averages, $(37 + 46)/2$. **C. 42** takes the overall average for the middle number.

D. 47 treats the groups as five numbers each: $462 - (5 \times 37 + 5 \times 46) = 47$.

Q4**ANSWER C**

Confidence: Pattern 48% | Exact-question 11%

Each term is twice the one before it plus a number that grows by one each time:

$3 \times 2 + 1 = 7$, $7 \times 2 + 2 = 16$, $16 \times 2 + 3 = 35$, $35 \times 2 + 4 = 74$, and so $74 \times 2 + 5 = 153$.

The differences 4, 9, 19, 39 confirm it: each is twice the one before plus 1, so the next is 79 and $74 + 79 = 153$.

A. 113 adds the last difference, 39, again instead of the next one. **B. 148** only doubles and forgets the added number. **D. 154** adds 6, skipping 5.

Q5**ANSWER A**

Confidence: Pattern 48% | Exact-question 11%

A plane mirror swaps left and right, so every hand is reflected about the 12–6 line. The image time and the actual time therefore add to 12 hours. Writing 12:00 as 11:60,

$11:60 - 3:40 = 8:20$.

B. 8:40 reflects the hour hand but leaves the minute hand at 40. **C. 9:20** reflects the two hands separately, $12 - 3 = 9$ and $60 - 40 = 20$, without borrowing the hour. **D. 9:40** subtracts only the hour.

Q6**ANSWER D**

Confidence: Pattern 48% | Exact-question 11%

Revenue is energy $\times$ tariff. With 1 GWh = 10^6 kWh, the revenue in million rupees is the product of the two columns. Total energy is 460 GWh and total revenue is 2480 million rupees.

Category	Revenue (million rupees)	Share of energy	Share of revenue	Difference (points)
Domestic	540	26.09 per cent	21.77 per cent	-4.31
Commercial	480	13.04 per cent	19.35 per cent	+6.31
Industrial	1300	43.48 per cent	52.42 per cent	+8.94
Agricultural	160	17.39 per cent	6.45 per cent	-10.94

The largest excess of revenue share over energy share is **Industrial**, at 8.94 points.

B. Commercial has the highest tariff, but it is a small category, so its excess is only 6.31 points.

C. Agricultural has the largest gap of all, but in the wrong direction: its revenue share falls short of its energy share.

A. Domestic also pays less than the average tariff of $2480/460 = 5.39$ rupees per kWh, so its revenue share is below its energy share.

Q7**ANSWER B**

Confidence: Pattern 48% | Exact-question 11%

Count by the number of engineers on the committee.

Engineers	Managers	Committees
2	2	$C(5, 2) \times C(4, 2) = 10 \times 6 = 60$
3	1	$C(5, 3) \times C(4, 1) = 10 \times 4 = 40$
4	0	$C(5, 4) = 5$

Total **105** (confirmed by enumerating all 126 committees).

A. 60 counts only the committees with exactly two engineers. **C. 125** answers "at least one engineer": all $C(9, 4) = 126$ committees less the single all-manager one. **D. 126** is every committee, with no condition applied.

Q8**ANSWER A**

Confidence: Pattern 48% | Exact-question 11%

The passage says condition-based maintenance wastes less of a component's life, *but* that where the measurements cannot be relied upon a fixed interval is safer. Its advantage therefore holds only where the measurements can be trusted, which is **(A)**.

B. is about cost, which the passage never mentions; it speaks only of wasted useful life. **C.** reverses the passage: replacing a part whether or not it is worn is exactly what wastes useful life.

D. goes beyond the passage, which says sensors are needed for the second approach and that they may be unreliable, not that fitting them settles the matter.

Q9**ANSWER D**

Confidence: Pattern 48% | Exact-question 11%

Section 15 — The Last-Minute Sheet

What follows is the set of formulae, constants and rules of thumb worth carrying into the last day, grouped by the ten technical sections of the 2027 syllabus. Each group opens with the syllabus's own text for that section, quoted as printed, so that the sheet can be checked against what the paper may ask: every topic the syllabus names has an entry here, including the ones no counted paper has yet examined. Every numerical example on the sheet was recomputed before printing.

Two things this sheet deliberately is not. **It is not a syllabus course.** A formula list cannot tell you what a formula is for, and a candidate meeting one of these for the first time on this sheet is meeting it too late. **And it does not replace the chapters.** Each entry is the short form of something the subject chapter works through in full; where a result has a condition — a linear element, a lossless line, a continuous inductor current, a stable closed loop, an ideal op-amp — the condition is printed beside it, because a result used without its condition is the commonest wrong answer on the paper.

Notation: $j = \sqrt{-1}$; $\omega = 2\pi f$; phasors and AC meter quantities are RMS unless marked "peak" or with a subscript m; three-phase voltages are line-to-line unless stated; $\ln$ is the natural logarithm and $\log_{10}$ is written out; $R_1 || R_2 = R_1 R_2 / (R_1 + R_2)$; pu is per unit; δ in a machine or stability formula is the electrical load angle.

Constants and numbers to have ready

Item	Value
Permittivity of free space	$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$; $1/(4\pi\epsilon_0) \approx 8.988 \times 10^9 \text{ m/F}$ (use 9×10^9)
Permeability of free space	$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$
Speed of light	$c = 1/\sqrt{\mu_0\epsilon_0} \approx 3 \times 10^8 \text{ m/s}$
Electron charge, Boltzmann constant	$e = 1.602 \times 10^{-19} \text{ C}$; $k = 1.381 \times 10^{-23} \text{ J/K}$
Thermal voltage	$V_T = kT/q \approx 25.9 \text{ mV}$ at 300 K (26 mV is the usual round figure)
Gravity, water	$g = 9.81 \text{ m/s}^2$; density of water 1000 kg/m^3
Surds and ratios	$\sqrt{2} = 1.4142$, $\sqrt{3} = 1.7321$, $1/\sqrt{2} = 0.7071$, $2/\pi = 0.6366$, $2\sqrt{2}/\pi = 0.9003$, $3\sqrt{2}/\pi = 1.3505$, $4/\pi = 1.2732$, $3/\pi = 0.9549$
Logarithms	$\ln 2 = 0.6931$, $\ln 10 = 2.3026$; $20 \log_{10} 2 = 6.02 \text{ dB}$; $20 \log_{10} \sqrt{2} = 3.01 \text{ dB}$; a factor of 10 is 20 dB in voltage, 10 dB in power
The operator a	$a = 1\angle 120^\circ = -0.5 + j0.866$; $a^2 = 1\angle 240^\circ$; $1 + a + a^2 = 0$; $a^3 = 1$

1 Engineering Mathematics

Linear algebra

Syllabus — Linear Algebra. Matrix Algebra, Systems of linear equations, Eigen values, Eigen vectors.

Item	Result
Transpose, inverse	$(AB)^T = B^T A^T$; $(AB)^{-1} = B^{-1} A^{-1}$; 2×2 : $[[a, b], [c, d]]^{-1} = (1/(ad - bc)) [[d, -b], [-c, a]]$; $A^{-1} = \text{adj } A / \det A$
Determinant	$\det(AB) = \det A \det B$; $\det A^T = \det A$; $\det(kA) = k^n \det A$ for $n \times n$; $\det A^{-1} = 1/\det A$; triangular: product of the diagonal; a row swap changes the sign; two equal rows give 0

Item	Result
Special matrices	symmetric $A^T = A$; skew-symmetric $A^T = -A$ (zero diagonal, $x^T A x = 0$ for every x , eigenvalues 0 or purely imaginary); orthogonal $A^T A = I$ ($\det = \pm 1$, every $ \lambda = 1$, preserves lengths and dot products); idempotent $A^2 = A$ (eigenvalues 0 and 1)
Rank	number of independent rows = number of independent columns = order of the largest non-zero minor; $\text{rank } A = \text{rank } A^T$; $\text{rank}(AB) \leq \min(\text{rank } A, \text{rank } B)$
Linear dependence	more than n vectors in $\mathbb{R}^n$ are always dependent; n vectors in $\mathbb{R}^n$ are independent iff the matrix they form has $\det \neq 0$
Systems $Ax = b$, n unknowns	unique solution iff $\text{rank } A = \text{rank } [A \mid b] = n$; infinitely many iff $\text{rank } A = \text{rank } [A \mid b] < n$; none iff $\text{rank } A < \text{rank } [A \mid b]$. Square A : unique iff $\det A \neq 0$, and then $Ax = 0$ has only $x = 0$
Homogeneous $Ax = 0$	always consistent; $n - \text{rank } A$ free parameters; non-trivial solution iff $\text{rank } A < n$
Eigenvalues	$\det(A - \lambda I) = 0$; $\sum \lambda = \text{trace } A$, $\prod \lambda = \det A$; 2×2 : $\lambda^2 - (\text{trace})\lambda + \det = 0$
Spectral mapping	if $Av = \lambda v$ then A^k has λ^k , A^{-1} has $1/\lambda$, $A + cI$ has $\lambda + c$, $p(A)$ has $p(\lambda)$, e^A has e^λ — all with the same eigenvector v
Special spectra	triangular or diagonal: the diagonal entries; real symmetric: real eigenvalues, and eigenvectors of distinct eigenvalues are orthogonal
Eigenvectors	eigenvectors of distinct eigenvalues are independent; an $n \times n$ matrix with n distinct eigenvalues is diagonalisable, $A = PDP^{-1}$
Cayley–Hamilton	every square matrix satisfies its characteristic equation. $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$: $A^2 - 5A - 2I = 0$, so $A^{-1} = (A - 5I)/2 = \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix}$
Plane and normal	the plane $w^T x = c$ has normal vector w ; distance from the origin $ c / w $
2-D rotation by θ	$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$: orthogonal, $\det 1$, eigenvalues $e^{\pm j\theta}$

Calculus

Syllabus — Calculus. Mean value theorems, Theorems of integral calculus, Evaluation of definite and improper integrals, Partial Derivatives, Maxima and minima, Multiple integrals, Fourier series, Vector identities, Directional derivatives, Line integral, Surface integral, Volume integral, Stokes's theorem, Gauss's theorem, Divergence theorem, Green's theorem.

Item	Result
Rolle	f continuous on $[a, b]$, differentiable on (a, b) , $f(a) = f(b) \Rightarrow f'(c) = 0$ for some c in (a, b)
Lagrange mean value theorem	$f'(c) = (f(b) - f(a))/(b - a)$ for some c in (a, b)
Cauchy mean value theorem	$(f(b) - f(a))/(g(b) - g(a)) = f'(c)/g'(c)$, $g' \neq 0$ on (a, b)
Fundamental theorem	$d/dx \int_a^x f(t) dt = f(x)$; so $F(x) = \int_0^x f(t) dt$ increases where $f > 0$ and decreases where $f < 0$
Leibniz rule	$d/dx \int_{a(x)}^{b(x)} f(t) dt = f(b(x)) b'(x) - f(a(x)) a'(x)$
Definite-integral properties	$\int_0^a f(x) dx = \int_0^a f(a - x) dx$; $\int$ from $-a$ to a is $2 \int_0^a f$ for even f and 0 for odd f ; over $(0, \infty)$, the substitution $x \rightarrow 1/x$ often turns an integral into a copy of itself
Standard integrals	$\int_0^\infty x^n e^{-ax} dx = n!/a^{n+1}$; $\int_0^\infty e^{-ax^2} dx = \frac{1}{2}\sqrt{\pi/a}$; $\int_0^\infty dx/(1+x^2) = \pi/2$; $\Gamma(n+1) = n!$, $\Gamma(1/2) = \sqrt{\pi}$
Wallis	$\int_0^{\pi/2} \sin^n x dx = \int_0^{\pi/2} \cos^n x dx = [(n-1)(n-3)\dots]/[n(n-2)\dots]$, multiplied by $\pi/2$ when n is even
Improper integrals	$\int_1^\infty dx/x^p$ converges iff $p > 1$; $\int_0^1 dx/x^p$ converges iff $p < 1$; test an infinite limit or an infinite integrand by comparison
Partial derivatives	$f_{xy} = f_{yx}$ when both are continuous; chain rule $df/dt = f_x dx/dt + f_y dy/dt$; Euler's theorem for f homogeneous of degree n : $x f_x + y f_y = n f$

Section 16 — Sources, Conventions and Limits

This chapter says what the book was built from, how the counting was done, and what the book does not claim. It is here so that a reader who doubts a figure knows which file to open.

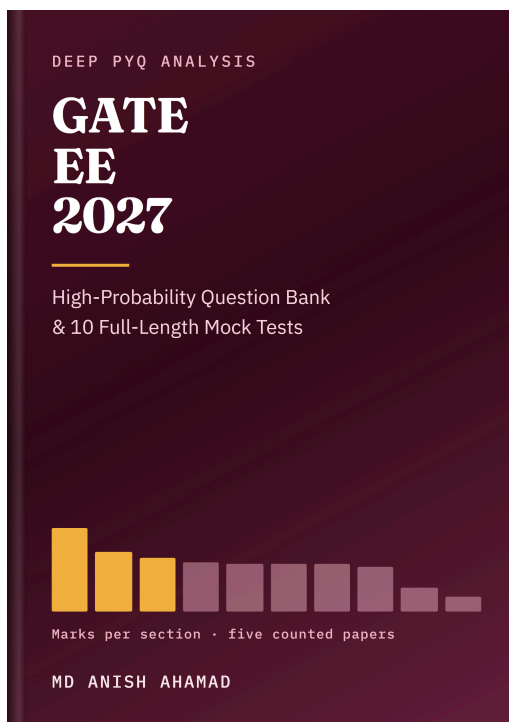
What the book was built from

Four official syllabus documents. The GATE 2026 Electrical Engineering syllabus, published by IIT Guwahati, and the GATE 2027 syllabus, published by IIT Madras, both held as PDFs in [research/official/](#). Section 2 reproduces the 2027 document and compares the two phrase by phrase and token by token; the comparison is produced by a program that reads the two PDFs, not from memory, and it finds no phrase added, removed or reworded — only one corrected spelling, "Butterwoth" in 2026 and "Butterworth" in 2027, and a change in how the section names are laid out on the page. The 2024 (IISc Bengaluru) and 2025 (IIT Roorkee) EE syllabi, also held in [research/official/](#), were compared with 2027's the same way: they name the same 217 phrases, and differ from it in that one spelling and, for 2024, in fourteen changes of capitalisation or punctuation. The 2022 and 2023 syllabi could not be obtained: the 2022 organiser's GATE site (IIT Kharagpur) and the 2023 organiser's (IIT Kanpur) could not be reached on the download date, and no copy from any other site was used in their place. This book states nothing about those two documents.

Five official question papers and their five official answer keys — the EE papers of 2022 (organised by IIT Kharagpur), 2023 (IIT Kanpur), 2024 (IISc Bengaluru), 2025 (IIT Roorkee) and 2026 (IIT Guwahati): 325 questions and 500 marks. EE is set as one paper a year, so a citation in this book names a year and the printed question number, as in "2025 (Q35)". Every file was downloaded from a GATE organising-institute website. No coaching site, aggregator, question repository or blog was used for any paper, key, syllabus or figure in this book.

File	Source
2022 question paper	https://gate2026.iitg.ac.in/doc/download/2022/ee_2022.pdf
2022 answer key	https://gate2026.iitg.ac.in/doc/download/Answer_keys2022/ee_2022.pdf
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2026 answer key	https://gate2026.iitg.ac.in/doc/download/2026/Keys/EE_Keys.pdf
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2026 syllabus	https://gate2026.iitg.ac.in/doc/GATE2026_Syllabus/EE_2026_Syllabus.pdf
2027 syllabus	https://gate2027.iitm.ac.in/static/doc/GATE2027_Syllabus/EE_GATE2027_Syllabus.pdf

All fourteen were downloaded on 30 September 2026. The 2022, 2023 and 2024 papers and keys come from the previous-years archive on the GATE 2026 (IIT Guwahati) download page, which lists exactly one EE paper and one EE key for each of those years; the examinations themselves were organised by IIT Kharagpur, IIT Kanpur and IISc Bengaluru. The same archive's copy of the 2025 paper is byte-identical to the IIT Roorkee copy listed above.



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